

EVA SINGH

Iowa City, IA | +1 319 936 1467 | evasinghwork@gmail.com | linkedin.com/in/eva-singh295

SUMMARY

Software engineer experienced in building full-stack applications, real-time systems, AI-powered tools, and data pipelines. Skilled in APIs, distributed systems, and interactive applications using C++, Python, Java, React, Flask, and modern cloud tooling. Interested in high-performance software systems, real-time data infrastructure, and applied AI.

EDUCATION

University of Iowa

B.S.E. in Computer Science and Engineering

Iowa City, IA

Minor in Artificial Intelligence | Certificate in Modeling and Simulation

Expected May 2027

Cumulative GPA: 3.47/4.0

Awards: International Distinction in Education Award; Dean's List (2024–2026); 1st Place, IEEE Region 4 Ethics Competition (2026); Emerging Leader Award (2025); Campus Leadership Award (2026); Academics Award (2026)

Relevant Coursework: Embedded Systems, Algorithms; Software Design; Computer Architecture and Organization; Digital Design; Probability & Statistics for Engineering; AI & Machine Learning in Engineering; Artificial Intelligence Fundamentals; Generative AI Tools; Independent Research in Electrical Engineering

TECHNICAL SKILLS

Languages & Frameworks: Python, PyTorch, Java, C++, C#, SQL, TypeScript, Vue 3, Flask, React, Socket.IO

Tools & Platforms: Git, Docker, AWS ECR, MongoDB, Unity, Postman, VS Code, IntelliJ IDEA, PyCharm, CLion

Machine Learning & AI: scikit-learn, Weights & Biases, MLflow, Jupyter, NumPy, pandas, GANs (WGANs), Deep Learning, NLP, Sentiment Analysis, OpenAI API

Professional: Stakeholder Management, Cross-Functional Collaboration, Technical Communication, Project Management, Mentorship & Coaching, Strategic Planning

ENGINEERING EXPERIENCE

Software Engineering Intern | Musco Sports Lighting

May 2025 – Present

- Built and maintained two real-time strike zone systems: a Vue.js interface used daily by internal and external teams to test and view live data, and a Python/MQTT-based system for live sports broadcasts, expanded to a second deployment site
- Analyzed and visualized 100,000+ telemetry records via Vue.js dashboards, enabling stakeholders to make data-driven product decisions
- Engineered Flask and Socket.IO APIs for real-time data routing and low-latency streaming
- Authored database queries and tested 25+ API endpoints in Postman to validate reliability and performance
- Modeled and improved Python-based game calculation logic by 40%, processing large-scale JSON datasets for sports data
- Containerized services with Docker and deployed via AWS ECR for consistent environments across systems
- Led development of “MuscoBoard,” a live scoreboard system built from scratch, winning a department-wide hackathon among 6 teams under a tight deadline
- Evaluated text-to-speech models for emerging sports technology applications

Research Assistant | Advanced Pulmonary Physiomic Imaging Laboratory, UI

Aug 2025 – Dec 2025

- Trained Wasserstein GANs (PyTorch) for medical image generation and generative modeling research
- Evaluated model stability and output quality using statistical metrics and visualization tools
- Conducted hyperparameter tuning and experiment tracking with Weights & Biases and MLflow in Jupyter Notebooks

Web Intern | Marketing & Communications, College of Engineering, UI

Aug 2024 – Present

- Build and maintain College of Engineering websites using SiteNow CMS and structured templates
- Develop accessible, WCAG-compliant web pages for academic programs
- Monitor analytics, resolve support tickets, and debug frontend/backend performance issues
- Optimize layouts, permissions, and content structure across departmental sites

Research Volunteer | Holo Reality Lab, University of Iowa

Jan 2025 – Mar 2025

- Developed backend systems in C# for a Unity-based virtual reality teaching application
- Integrated core components to support immersive, adaptive simulation environments

- Collaborated with a multidisciplinary team to enhance functionality and user experience

PROJECTS

Latent Lyrics (LLM Music Analysis Platform) | *Python, React, OpenAI/Spotify APIs* December 2025

- Led a team of 4 building an app that analyzes song lyrics in real time using LLMs, integrating Spotify, Genius, and OpenAI APIs across 100+ languages
- Engineered real-time lyric-to-playback syncing, batching one request per song to minimize latency
- Built an AI-generated “vibe board” image feature per song, structuring model outputs for reliable parsing

Silly Goose Club (Hackathon Game) | *Java, TCP Sockets* December 2025

- Built and tested a multiplayer game using Java and multithreaded TCP client-server architecture, managing concurrent game state across synchronized rooms
- Won 1st place in a software design course hackathon for the completed application and presentation, led by me

LEADERSHIP & CAMPUS INVOLVEMENT

President | *IEEE At Iowa, UI* May 2026 – Present

- Lead a 50+ member organization and a 5-person executive board, setting priorities and overseeing planning and funding
- Won 1st place at the IEEE Region 4 Nexus Ethics Case Study Competition, analyzing a real-world ethics dilemma and presenting a solution to a judging panel

Vice President External | *Society of Women Engineers, UI* May 2025 – Present

- Lead corporate outreach and manage partnerships with companies including Collins Aerospace, John Deere, and Emerson
- Coordinate engagement and professional development initiatives for 200+ members
- Serve as liaison between students and industry partners for recruiting and events

Professional Development Chair | *Society of Women Engineers, UI* Aug 2024 – May 2025

- Founded and led SWE's first Professional Development Week, engaging 100+ students in career-focused programming
- Partnered with NSBE, SHPE, and 6+ companies to organize networking and employer engagement events attended by college leadership
- Planned workshops, panels, and recruiting events supporting internship and career readiness outcomes

Peer Mentor | *iPeer Program, ISSS, UI* Jun 2025 – Dec 2025

- Mentored incoming international students through one-on-one check-ins on academics, campus resources, and adjustment to university life
- Fostered belonging and confidence by connecting mentees to involvement opportunities and support services

International Student Ambassador | *Admissions, UI* Sep 2023 – Present

- Serve as a point of contact for prospective international students and families on academics, student life, and enrollment
- Selected to be featured on the University of Iowa Admissions website as an ambassador profile

Web Content Director | *Debate Club at Iowa* Sep 2023 – May 2024

- Managed Instagram and LinkedIn presence, expanding digital engagement
- Developed content strategy promoting events, achievements, and educational resources

INTERESTS

Freshwater fishing | Perfecting homemade biryani recipes